Albert Ankomah Dodoo

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SUMMARY

I am a results-driven Machine Learning researcher with expertise in computer vision, pattern recognition, swarm intelligence, and optimization algorithms. I am skilled in using machine learning models such as Vision Transformers (ViT), Generative Adversarial Networks, Graph Neural Networks (GNNs), and Long Short-Term Memory (LSTM) for application in healthcare, cybersecurity, ISRU, and robotics. I am proficient in using research findings from in-depth analysis to support decision-making. I have a proven track record in mentoring undergraduate and master's students interested in machine learning research and its application.

RESEARCH INTEREST

- Computer Vision
- Intrusion and Anomaly Detection
- Pattern Discovery and Time Series Analysis

EDUCATION

University of Ghana Accra, Ghana
MSc. Computer Science Sep 2024 – Sep 2025
Thesis: Adaptive Memory-Based Cuckoo Search for High-Dimensional Feature Optimization in Non-Stationary Fraud Detection Systems.

University of Ghana Accra, Ghana
BSc. Computer Science Aug 2019 – Sep 2023
Thesis: Resampled Correlation-Based Feature Descriptors: A Novel Approach to Enhancing Malware Detection Capabilities.

WORK EXPERIENCE

University of Ghana Department of Computer Science Accra, Ghana
Graduate Assistant Oct 2025 – Present

- Support academic writing by converting thesis documents into publishable research articles.
- Organise and deliver biweekly LaTeX workshops for over 50 undergraduate and graduate final-year students.
- Assist in grading mid-semester assessments and final examinations for 120 graduate students.
- Conduct tutorial sessions in Advanced Data Structures and Algorithms, Intelligent Systems, and Natural Language Processing for 120 students.

Data Intelligence and Swarm Analytics Laboratory (DISAL) Accra, Ghana
Research Assistant Aug 2022 - Present

- Mentor 11 undergraduate and master's students in machine learning research, guiding projects through weekly meetings.
- Collaborated with two undergraduate students on conducting research on Vision Transformer-based COVID-19 detection using chest CT scans.
- Achieved 98% accuracy in malware detection by employing an LSTM model on correlation-based feature descriptors.
- Researched an adaptive memory-based Cuckoo Search algorithm for feature optimization in non-stationary fraud detection scenarios, addressing concept drift.

Knoxxi Ghana Limited Accra, Ghana
Data Scientist (Contract) Oct 2025 – Apr 2026

- Designed and optimized complex SQL queries (including multi-table JOINs) to transform raw health data into structured datasets for analysis and reporting.
- Analyzed large-scale patient vital datasets to generate actionable insights that support clinical and business decision-making.

- Developed interactive dashboards and visualizations using Apache Superset and Tableau to communicate trends, anomalies, and key performance indicators to stakeholders.

Space Copy Inc. Delaware, United States
STEM Research Intern Jun 2025 – Sep 2025

- Reviewed 50+ peer-reviewed papers on on-device AI, runtime optimization, and explainability for autonomous edge systems.
- Assessed embedded vision and sensor fusion methods in lunar conditions, focusing on optimization trade-offs between accuracy and efficiency.
- Supported grant writing for the NASA CAN on SLM Additive Manufacturing Dual Technologies (MSFC).

University of Ghana Department of Computer Science Accra, Ghana
Teaching and Research Assistant Sep 2023 – Oct 2024

- Conducted a systematic literature review on network security and privacy using the PRISMA framework to explore trends since 2020.
- Led tutorial sessions for 70 students on subjects like Machine Learning, Numerical Methods, and Statistical Models for Data Science to improve their understanding.
- Demonstrated practical applications of statistical concepts such as standardization, z-score, and measures of central tendency through Python code implementations during lectures.
- Designed quizzes for 70 students and provided feedback to enhance students' understanding.

LEADERSHIP & MENTORING

- African Computer Vision Summer School (ACVSS), Local Organiser
- Research Mentor, Data Intelligence and Swarm Analytics Laboratory (DISAL), 2022 – Present

PUBLICATIONS AND PREPRINTS

- Appati, K.J., Ziamah, B., Akrofi, H.A., & **Dodoo, A.A.** SARS Detection in Chest CT-Scan Images Using Bootstrapped ViT-B16 Model. Springer, Iran Journal of Computer Science (2025). doi.org/10.1007/s42044-025-00231-1
- **Dodoo, A.A.,** Koranteng, M., Amissah, D.K., & Appati, K.J. Resampled Correlation-Based Feature Descriptors: A Novel Approach to Enhancing Malware Detection Capabilities. Preprint at *Research Square* (2023). doi.org/10.21203/rs.3.rs-4219089/v1
- Quaicoo, R., Acheampong, R., Gyamenah, P., **Dodoo, A.A.,** Soli, M.A.T., & Appati, K.J. Adapting Triple BigGAN for Image Detection Tasks: Challenges and Opportunities. Preprint at *Research Square* (2023). doi.org/10.21203/rs.3.rs-4262097/v1
- Tetteh, L.M., Sarfo, O.Y., Amissah, D.K., **Dodoo, A.A.,** Soli, M.A.T., & Appati, K.J. (2023). Enhancing Unsupervised Signature Verification Through Advanced Feature Extraction Techniques. Preprint at *Research Square* (2023). doi.org/10.21203/rs.3.rs-4249016/v1

AWARDS, RECOGNITIONS, AND CERTIFICATIONS

- Overall Best Graduating Graduate Student at the Department of Computer Science (December 2025).
- DataCamp Donates X DataGlobal Hub Scholarship (October 2025)
- Overall Best Research Paper Award at the University of Ghana Postgraduate Students' Research Workshop (July 2025)
- Signal Processing (Python) for Neuroscience Practical Course, Udemy (January 2025).
- Large Language Models (LLM) Course, Vilnius University, Lithuania (December 2024).

TECHNICAL SKILLS

Programming Languages: Python | SAS | R | MATLAB | SQL
Machine Learning Models: CNN | LSTM | GRU | GNN | Autoencoders | Vision Transformers | GANs | Large Language Models
Libraries: TensorFlow | Keras | Scikit-learn | Pandas |NumPy | Pytorch | Hugging Face Transformers | NLTK | OpenCV
Visualization Tools: Apache Superset | Power BI
Productivity Tools: Git |Microsoft Office Suites | LaTeX